

“Calculus 3”

Multi-Variable Calculus

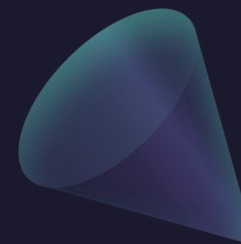
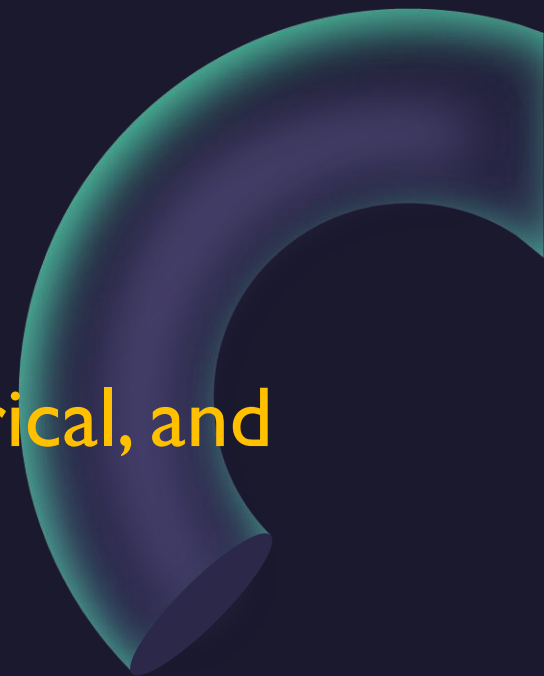
Instructor: Álvaro Lozano-Robledo

Day 15



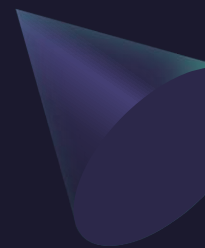
Any Reminders? Any Questions?

- Quiz on Friday on triple integrals in cartesian, cylindrical, and spherical coordinates.
- I will be away Wednesday-Thursday at a conference so no in-person class on 3/12 – watch videos instead!

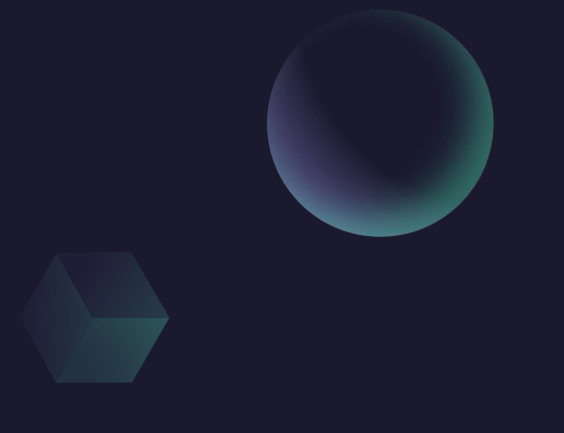




ALVARO: Start the recording!



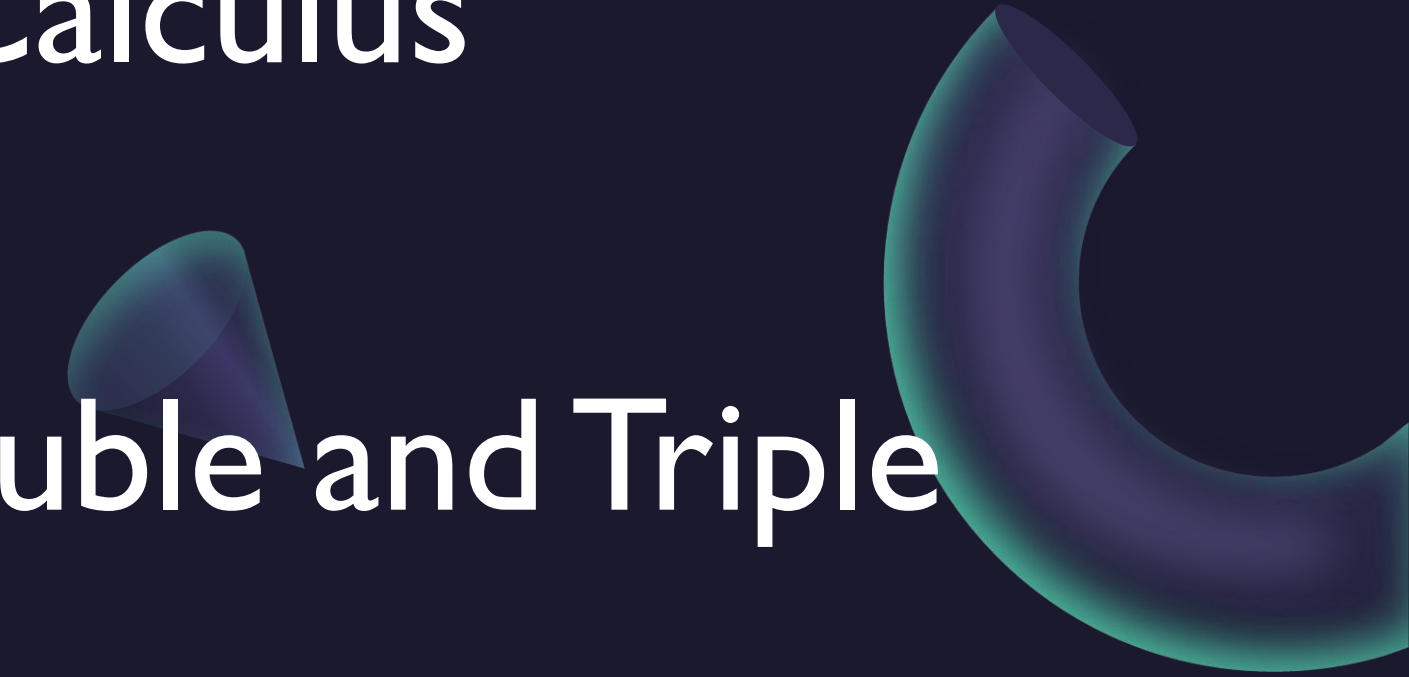
“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it, and the cone is further down and to the left.

Multi-Variable Calculus

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Exercises in Double and Triple Integrals

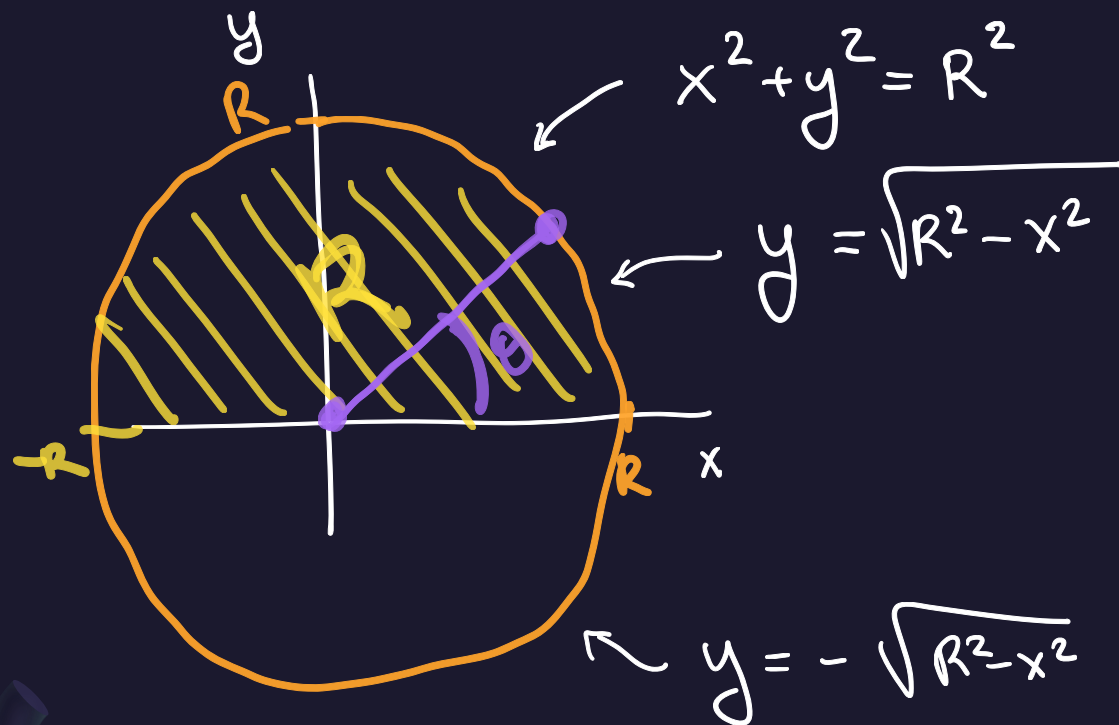
A cone and a torus are positioned in the lower right area of the slide. The cone is located above the torus, and both are partially obscured by the text of the second main heading.

Example:

T

F The area of the semi-circle centered at the origin with radius R , above the x -axis, can be computed with the following double integral:

$$\int_{-\pi/2}^{\pi/2} \int_0^R r \, dr \, d\theta.$$



$$\int_{-R}^R \int_{-\sqrt{R^2 - x^2}}^{\sqrt{R^2 - x^2}} 1 \, dy \, dx$$

$$R = \left\{ (r, \theta) : \begin{array}{l} 0 \leq r \leq R \\ 0 \leq \theta \leq \pi \end{array} \right\}$$

$$\iint_R 1 \, dA = \int_0^\pi \int_0^R 1 \cdot r \, dr \, d\theta$$

TRUE! THE AREA CAN BE COMPUTED W/ THAT INTEGRAL
BUT THE REGION DESC. IS THE WRONG SEMI CIRCLE!

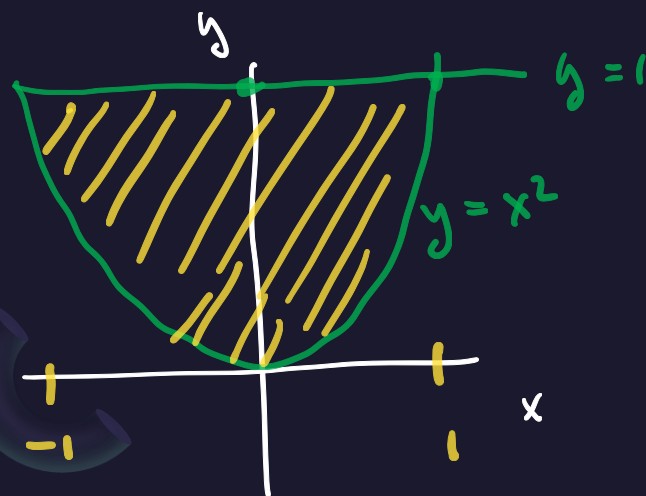
Example:

T F The following iterated integrals describe the volume of the same region E .

$$\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} 1 \, dz \, dy \, dx$$

$$\int_{-1}^1 \int_0^{1-x^2} \int_{x^2}^{1-z} 1 \, dy \, dz \, dx$$

$$A = \begin{cases} -1 \leq x \leq 1 \\ x^2 \leq y \leq 1 \\ 0 \leq z \leq 1-y \end{cases}$$



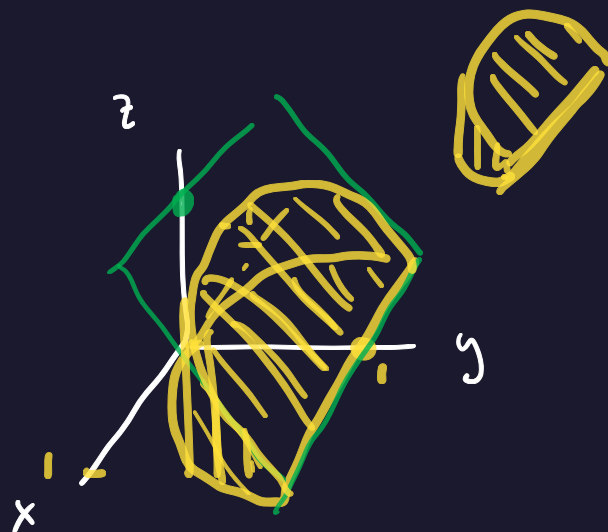
A

$$z = 1 - y$$

$$y + z = 1$$

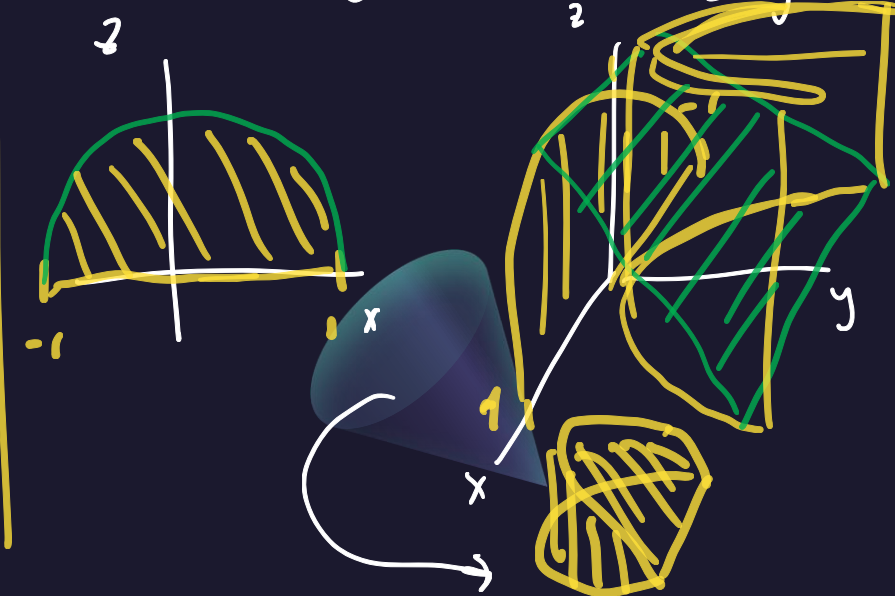
$$n = (0, 1, 1)$$

$$\begin{aligned} &\rightarrow (0, 1, 0) \\ &\sim (0, 0, 1) \\ &\sim (x, 1, 0) \end{aligned}$$



B

$$B = \begin{cases} -1 \leq x \leq 1 \\ 0 \leq z \leq 1 - x^2 \\ x^2 \leq y \leq 1 - z \end{cases}$$



Example:



F The following iterated integrals describe the volume of the same region E .

$$\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} 1 \, dz \, dy \, dx$$

$$\int_{-1}^1 \int_0^{1-x^2} \int_{x^2}^{1-z} 1 \, dy \, dz \, dx$$

$$A = \begin{cases} -1 \leq x \leq 1 \\ x^2 \leq y \leq 1 \\ 0 \leq z \leq 1-y \end{cases}$$

$$B = \begin{cases} -1 \leq x \leq 1 \\ 0 \leq z \leq 1-x^2 \\ x^2 \leq y \leq 1-z \end{cases}$$

$$B = \begin{cases} -1 \leq x \leq 1 \\ x^2 \leq y \leq 1 \\ 0 \leq z \leq 1-y \end{cases}$$

$$x^2 \leq y \leq 1-z \leq 1 \leadsto x^2 \leq y \leq 1$$

$$y \leq 1-z \Rightarrow z \leq 1-y$$

$$0 \leq z \leq 1-y$$

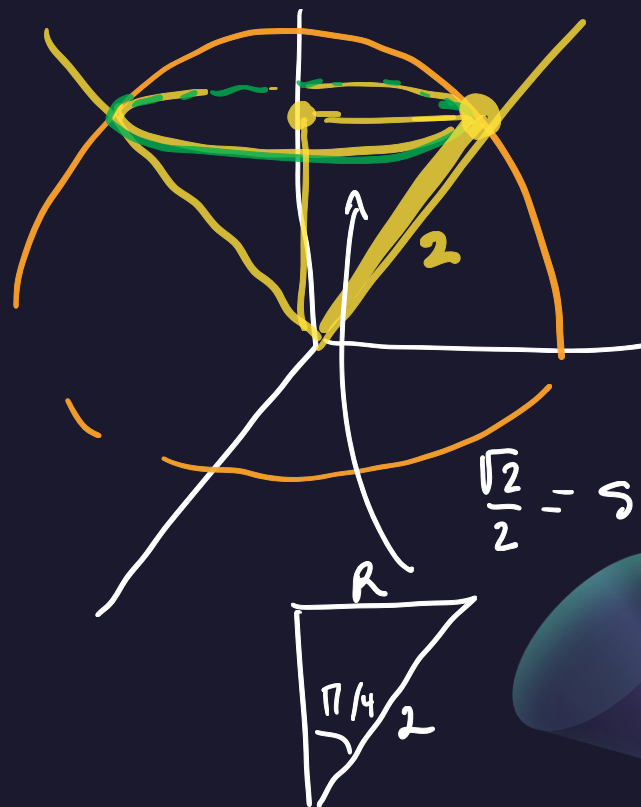
TRUE

Example:

The intersection of the surfaces given by $\phi = \frac{\pi}{4}$ and $\rho = 2$ is a circle of radius 2.

Spherical Coords: (ρ, θ, ϕ)

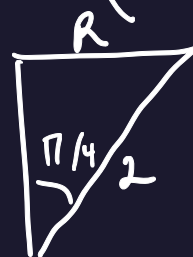
$$\begin{cases} \rho = 2 \\ \phi = \frac{\pi}{4} \end{cases}$$



yes
circle
BUT of
radius $\sqrt{2} \neq 2$

$$\frac{\sqrt{2}}{2} = \sin \frac{\pi}{4} = \frac{R}{2}$$

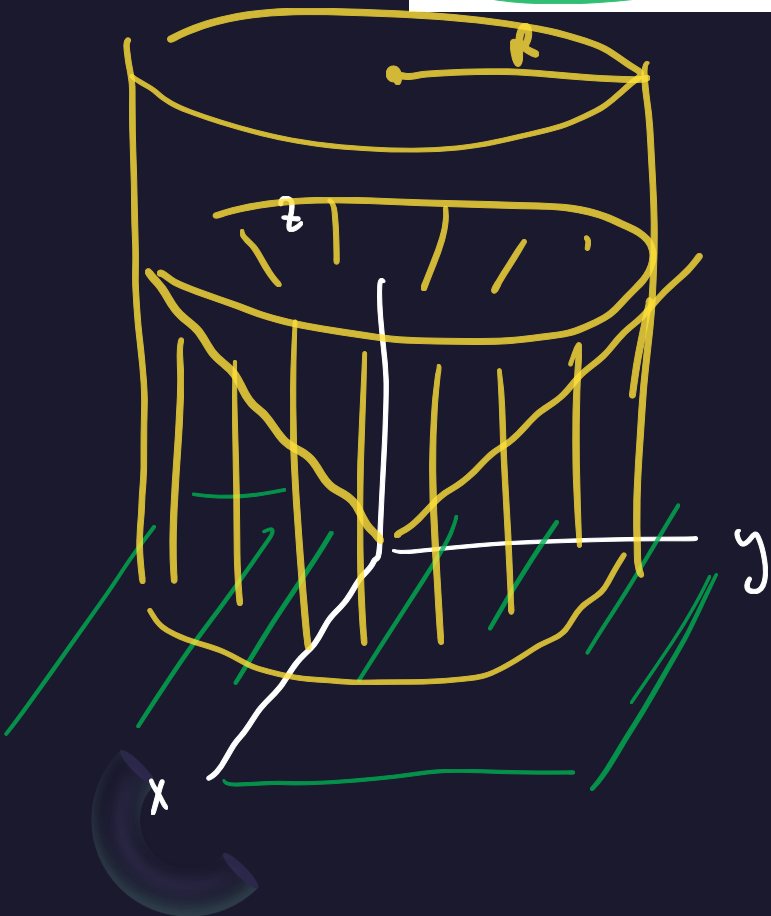
$$\Rightarrow R = \sqrt{2} \neq 2$$



Example:

Let E be the region above the xy -plane, below $z = \sqrt{x^2 + y^2}$, and inside $x^2 + y^2 = 9$. Write a triple integral below that represents the volume of E in two of the three coordinate systems that we have studied. For each part, circle or write the name of the coordinate system that you are using.

(a) Cartesian / Cylindrical / Spherical



CARTESIAN!

$$\begin{cases} -3 \leq x \leq 3 \\ -\sqrt{9-x^2} \leq y \leq \sqrt{9-x^2} \\ 0 \leq z \leq \sqrt{x^2+y^2} \end{cases}$$

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{\sqrt{x^2+y^2}} 1 \, dz \, dy \, dx$$

(CARTESIAN)

Example:

Let E be the region above the xy -plane, below $z = \sqrt{x^2 + y^2}$, and inside $x^2 + y^2 = 9$. Write a triple integral below that represents the volume of E in two of the three coordinate systems that we have studied. For each part, circle or write the name of the coordinate system that you are using.

(a) Cartesian / Cylindrical / Spherical

$$E = \begin{cases} 0 \leq r \leq 3 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq z \leq \sqrt{x^2 + y^2} = r \end{cases}$$

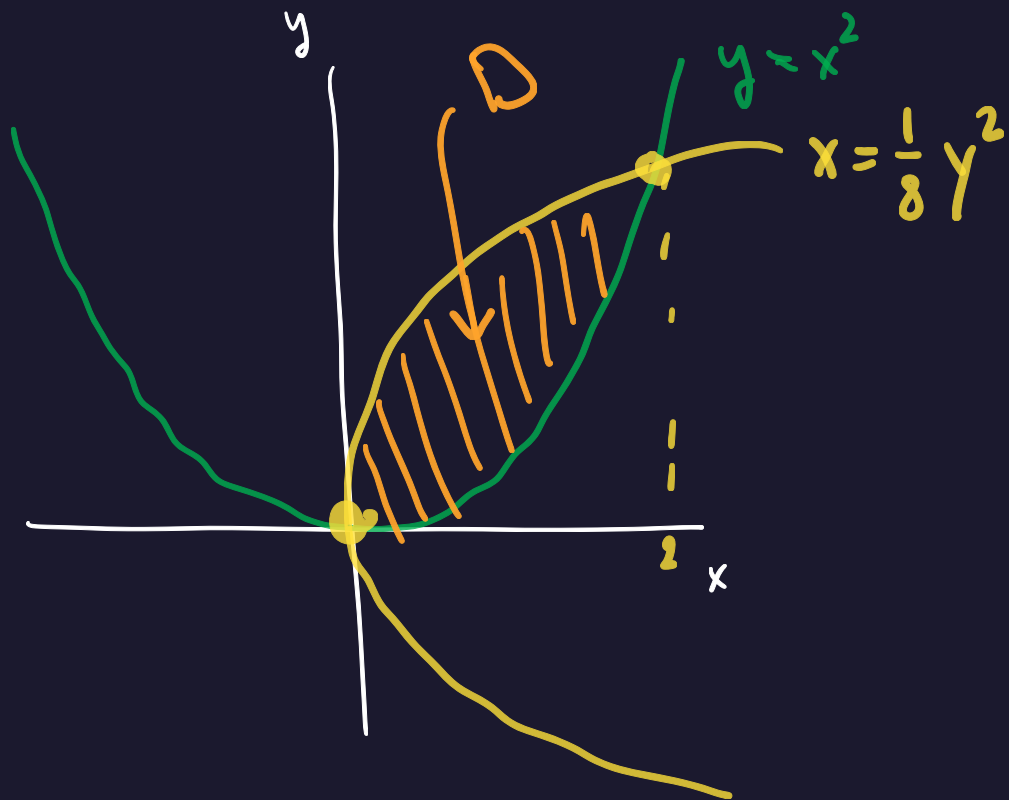
$$r^2 = x^2 + y^2$$

$$\begin{aligned} & \iiint_E 1 \, dV \\ &= \int_0^{2\pi} \int_0^3 \int_0^r 1 \cdot r \cdot dz \, dr \, d\theta \end{aligned}$$

Example:

Let D be the region in the xy -plane bounded by $y = x^2$ and $x = \frac{1}{8}y^2$.

- (a) Sketch the region D . Clearly indicate the region by shading and make sure to label the equations of any boundary curves.



$$\begin{cases} y = x^2 \\ x = \frac{1}{8} y^2 \end{cases}$$

$$x = \frac{1}{8} y^2 = \frac{1}{8} x^4$$

$$\frac{1}{8} x^4 - x = 0$$

$$x \left(\frac{1}{8} x^3 - 1 \right) = 0$$

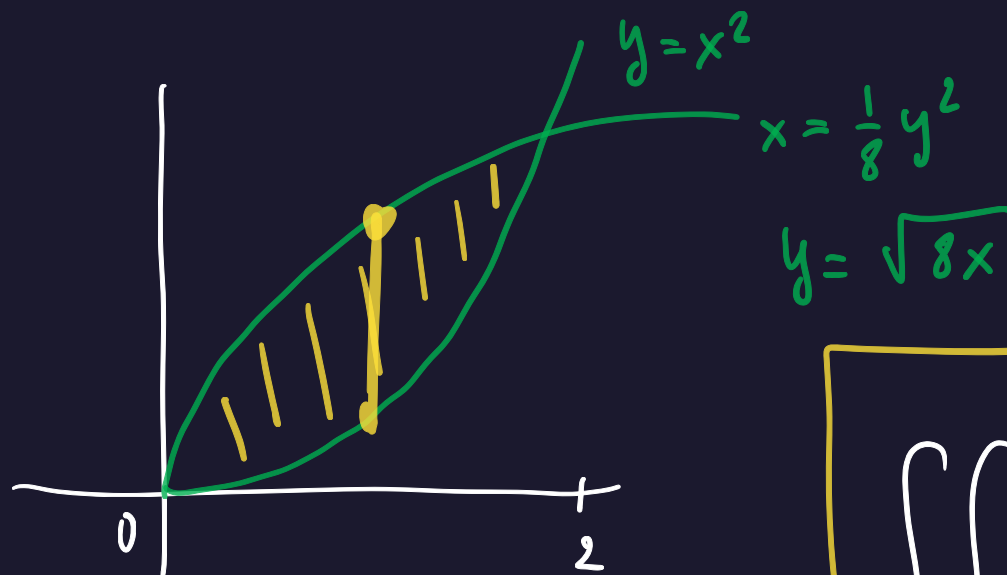
$$\begin{cases} x = 0 \\ \frac{1}{8} x^3 - 1 = 0 \end{cases}$$

$$\frac{1}{8} x^3 = 1$$

$$x^3 = 8 \quad |x=2|$$

Example:

(b) Write a double integral that represents the area of the region D in the order $dy dx$.

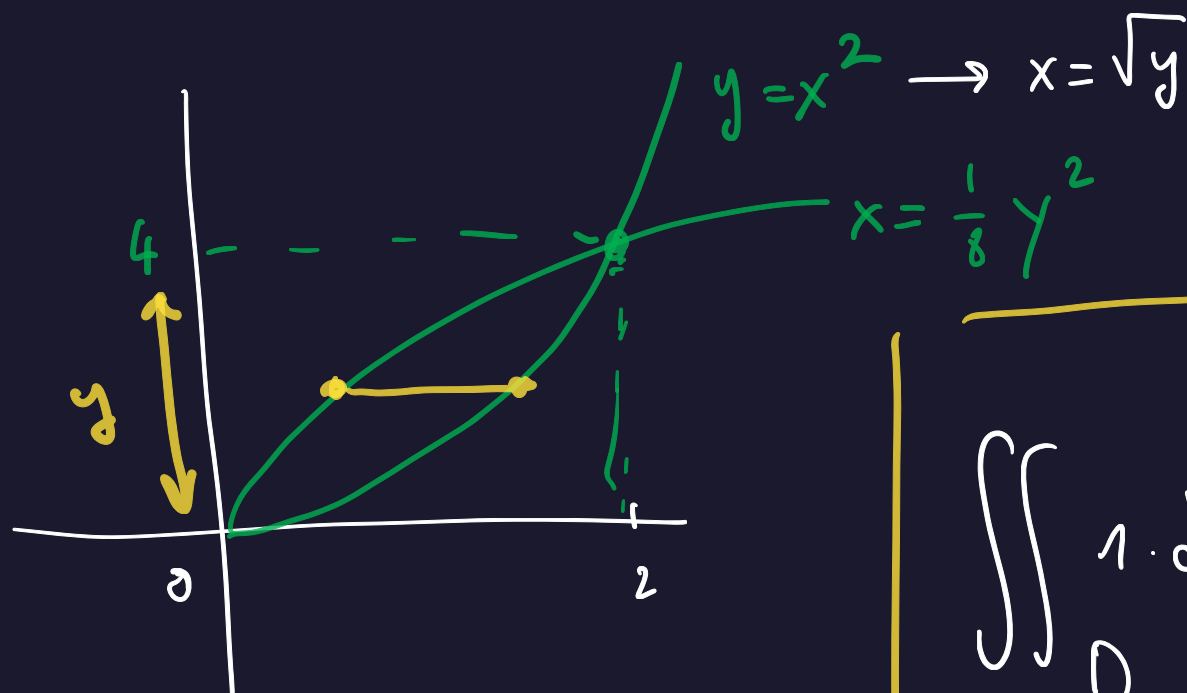


$$D = \begin{cases} 0 \leq x \leq 2 \\ x^2 \leq y \leq \sqrt{8x} \end{cases}$$

$$\iint_D 1 \, dA = \int_0^2 \int_{x^2}^{\sqrt{8x}} 1 \, dy \, dx$$

Example:

(c) Write a double integral that represents the area of the region D in the order $dx dy$.



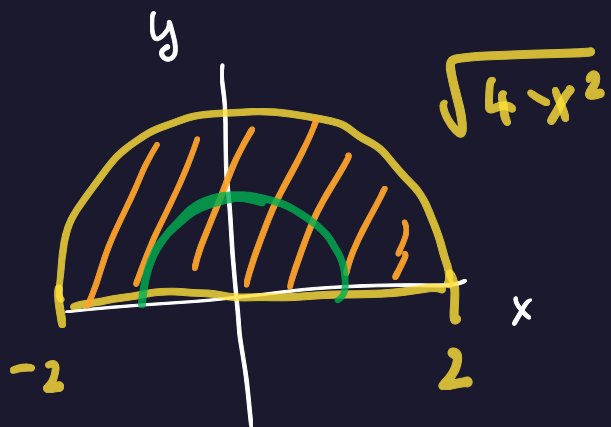
$$\iint_D 1 \cdot dA = \int_0^4 \int_{\frac{1}{8}y^2}^{\sqrt{y}} 1 \, dx \, dy$$

Example:

(a) Rewrite the following iterated integral using polar coordinates:

$$I = \int_{-2}^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2)^{3/2} dy dx.$$

(b) Evaluate the integral you wrote in part (a).



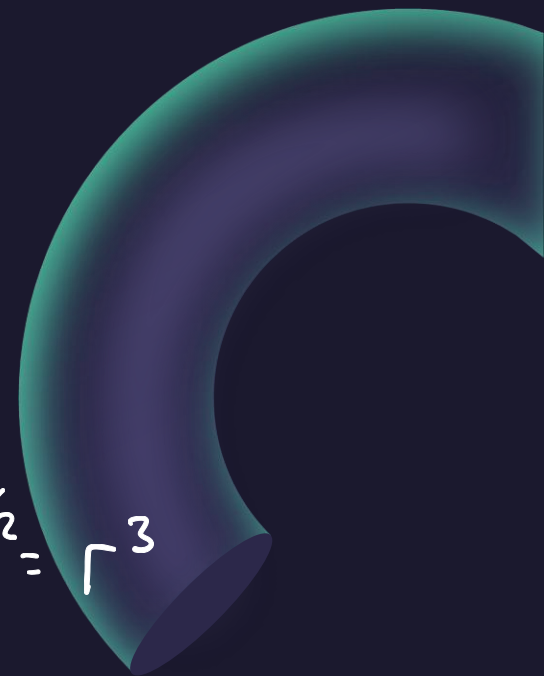
$$y = \sqrt{4-x^2}$$
$$\hookrightarrow y^2 = 4-x^2$$
$$x^2 + y^2 = 4$$

$$(x^2 + y^2)^{3/2} = (r^2)^{3/2} = r^3$$

$$I = \int_0^{\pi} \int_0^2$$

$$r^3 \cdot r dr d\theta =$$

$$\int_0^{\pi} \int_0^2 r^4 dr d\theta$$



Example:

(a) Rewrite the following iterated integral using polar coordinates:

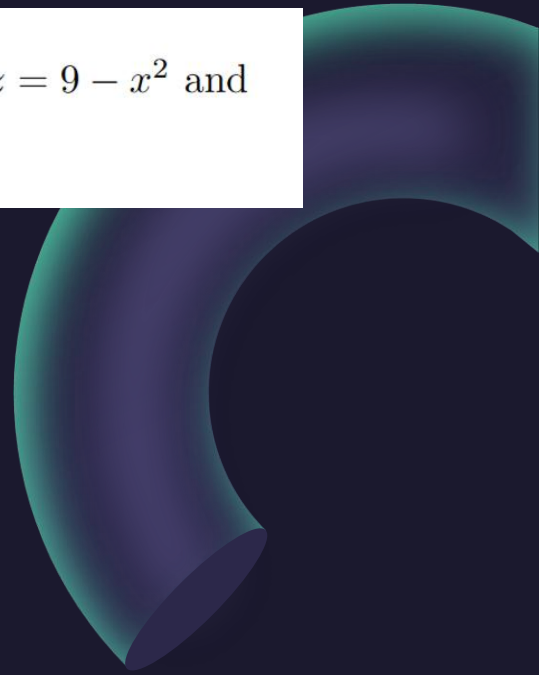
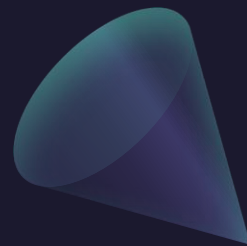
$$\int_{-2}^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2)^{3/2} dy dx.$$

(b) Evaluate the integral you wrote in part (a).

$$\begin{aligned} \int_0^{\pi} \int_0^2 r^4 dr d\theta &= \int_0^{\pi} \left. \frac{r^5}{5} \right|_0^2 d\theta = \int_0^{\pi} \frac{2^5}{5} d\theta \\ &= \frac{32}{5} \int_0^{\pi} d\theta = \frac{32}{5} \theta \Big|_0^{\pi} = \boxed{\frac{32\pi}{5}} \end{aligned}$$

Example:

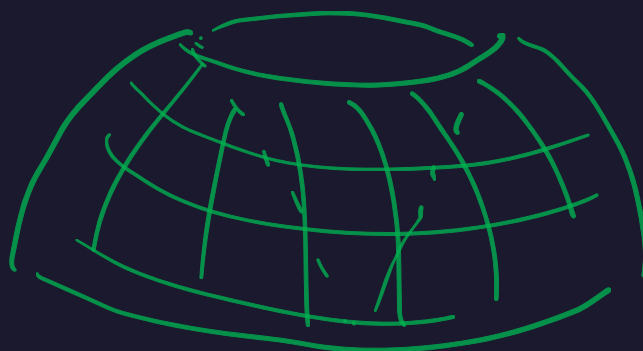
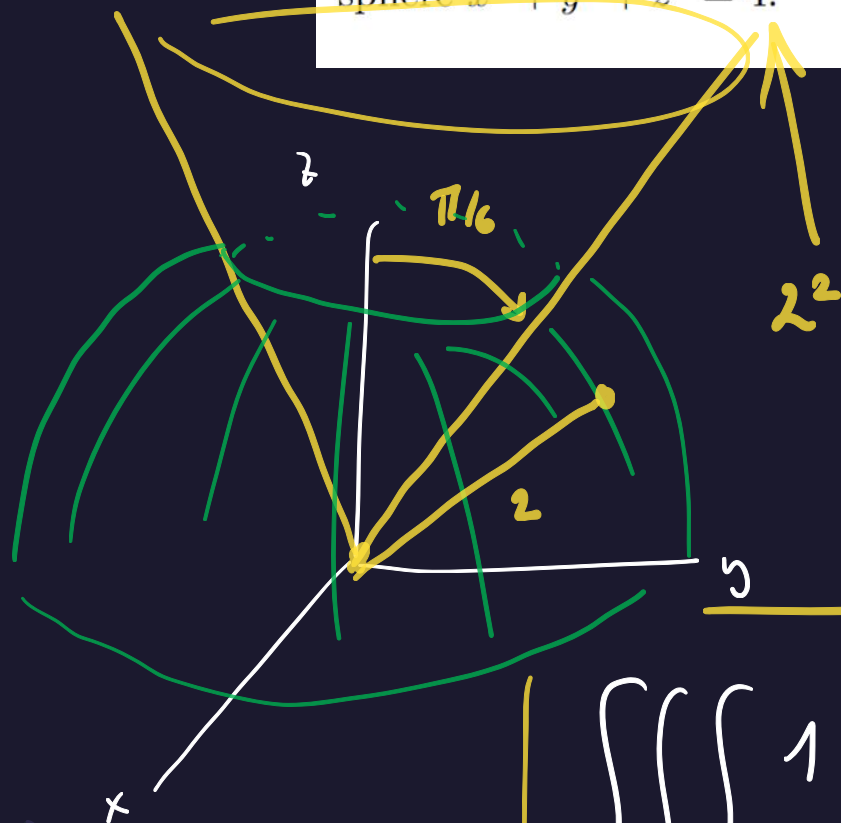
Set up the triple integral $\iiint_E x^2 e^y dV$, where E is the region enclosed by the surfaces $z = 9 - x^2$ and $z = 9 - y$ in the first octant. **Do not evaluate.**



Example:

Set up the following integral using either cylindrical or spherical coordinates. **Do not evaluate.**

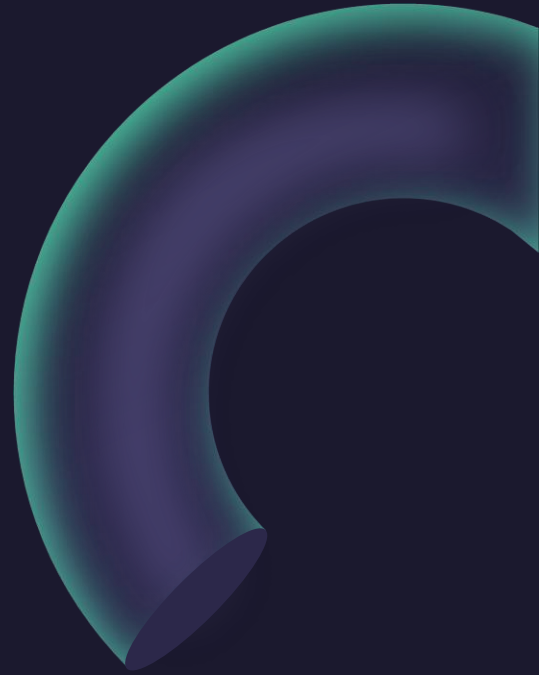
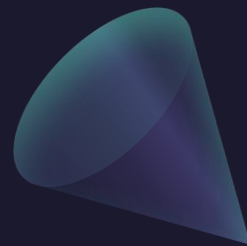
The volume of the region E , where E lies above the xy -plane, below the cone $\phi = \pi/6$ and inside of the sphere $x^2 + y^2 + z^2 = 4$.



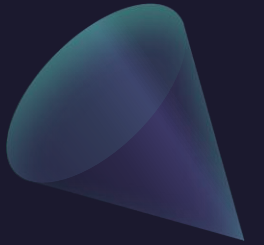
$$E = \left\{ (r, \theta, \phi) : \begin{array}{l} 0 \leq r \leq 2 \\ 0 \leq \theta \leq 2\pi \\ \pi/6 \leq \phi \leq \pi/2 \end{array} \right\}$$

$$\iiint_E 1 \, dV = \int_{\pi/6}^{\pi/2} \int_0^{2\pi} \int_0^2 1 \cdot r^2 \cdot \sin \phi \, dr \, d\theta \, d\phi$$

Example:



Questions?



Thank you

Until next time.

